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2 September 1987
SUPERSEDING
MIL-A-19083B(SHIPS)
11 March 1963
(See 6.7)

MILITARY SPECIFICATION

ACOUSTIC DEVICES, MINESWEEPING, A MARK 4(v) MOD 1 AND A MARK 4(v)NM

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers acoustic devices, A Mark 4(v) Mod 1 and A Mark 4(v)NM.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

TT-E-490	- Enamel, Silicone Alkyd Copolymer, Semigloss (For Exterior and Interior Use).
VV-G-632	- Grease, Industrial, General Purpose.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
DOD-P-15328	- Primer (Wash), Pretreatment (Formula No. 117 for Metals). (Metric)
MIL-P-15930	- Primer Coating, Shipboard, Vinyl-Zinc Chromate (Formula No. 120).
MIL-P-16298	- Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts and Associated Repair Parts.
MIL-I-17214	- Indicator, Permeability; Low-Mu (Go-No Go).
MIL-M-17413	- Motors, Direct Current, Integral H.P., Naval Shipboard.
MIL-S-17509	- Steel Castings, Austenitic, Chromium-Nickel, Low Magnetic Permeability.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-E-22200/2	- Electrodes, Welding, Covered (Austenitic Chromium-Nickel Steel).

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-278	- Welding and Casting Standard.

2.1.2 Other Government drawings and publication. The following other Government drawings and publication form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

S8101-A-1,412,161	- Small NM Plate.
S8101-B-1,464,365	- Identification Plate.
D-1,464,495	- Striker Cylinder.
B-3,103,978	- Seals.
B-3,103,979	- Motor Bed Plate.
B-3,103,980	- Base.

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NAVSEA DRAWINGS - (continued)

B-3,103,981 - Stiffner.
B-3,103,982 - End Plate.
B-3,103,983 - Clamp Bar.
B-3,103,985 - Gasket, Eccentric Ring.
B-3,103,986 - Cover Plate, Pillow Block.
B-3,103,987 - Eccentric Ring.
B-3,103,988 - Cylinder Sleeve.
B-3,103,990 - Cover Plate, Eccentric Ring.
B-3,103,992 - Socket Ring.
B-3,103,993 - Ball Socket.
B-3,103,994 - Snap Ring.
B-3,103,995 - Spacing ring.
B-3,103,996 - Gasket, Pillow Block.
B-3,103,997 - Stud Cylinder.
B-3,103,998 - Stud, Motor.
B-3,103,999 - Stud, Clamp Bar.
B-3,139,000 - Adjustment Bar.
B-3,139,002 - Stud, Bearing.
B-3,139,003 - Socket Ring Nut.
B-3,139,004 - Adjusting Screw.
B-3,139,005 - Staple.
B-3,139,006 - Wrench, Ring Nut.
B-3,139,007 - Guide Rail.
B-3,139,008 - Switch Post.
B-3,139,009 - Shield, Switch.
B-3,139,010 - Mounting Lug Block.
B-3,139,011 - Mounting Lug.
D-3,139,012 - Pillow Block.
C-3,139,013 - Side Rails.
B-3,139,014 - Tow Plate.
D-3,139,016 - Eccentric Shaft.
D-3,139,017 - Gasket, Diaphragm.
B-3,139,018 - Hanger Washer.
D-3,139,019 - Base Assembly.
D-3,139,021 - Striker.
D-3,139,022 - Spring.
H-3,139,023 - Dome Assembly Streamline Housing.
H-3,139,024 - Shell Assembly Streamline Housing.
H-3,139,025 - Tail Assembly Streamline Housing.
H-3,139,026 - Sound Box Assembly.
B-3,139,027 - Hanger Strip.
H-3,139,028 - Mechanism Assembly and Internal Wiring.
C-3,139,029 - Ball Stud.
B-3,139,030 - Cable Hanger.
B-3,139,032 - Electrical Receptacle Mounting Ring.
B-3,139,035 - Adj. Fork Block.
B-3,139,036 - Adj. Screw Fork.
B-3,139,037 - Adj. Screw Block.
B-3,139,042 - Cable Hanger Pad.
B-3,168,095 - Stiffner Pad.
B-3,168,096 - Support.
B-3,168,097 - Platform.

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NAVSEA DRAWINGS - (continued)

B-3,168,098 - Bolster Plate.
B-3,168,099 - Upper Doubler.
B-3,168,100 - Lower Doubler.
B-3,168,101 - Lug Bracket.
B-3,168,102 - Shell Clip, Forward.
B-3,168,103 - Clip Bracket, Forward.
B-3,168,104 - Doubler, Access.
B-3,168,105 - Fin.
B-3,168,106 - Fin Doubler.
B-3,168,107 - Vertical Stiffner.
B-3,168,108 - Horizontal Stiffner.
B-3,168,110 - Shell Clip, Aft.
B-3,168,111 - Clip Bracket, Aft.
C-3,168,112 - General Assembly A Mk 4(v) Mod 1, List of Materials.
C-3,168,113 - Cone, Sound Box.
D-3,168,114 - Bolting Ring.
D-3,168,115 - Diaphragm.
J-3,136,303 - Acoustic Device A Mk 4(v) Mod 1 General Assembly.
D-3,204,615 - Location of NM and Identification Plates.

PUBLICATION

NAVSEA
NAVSHIPS 250-530 - Operation and Maintenance of Mine Countermeasures Equipment.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN IRON AND STEEL INSTITUTE (AISI)
Steel Products Manual

(Application for copies should be addressed to the American Iron and Steel Institute, 350 Fifth Avenue, New York, NY 10017.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Material. Unless otherwise specified, material for fabrication of acoustic device A Mark 4(v) Mod 1, shall be as specified on drawings referenced in 2.1.2.

3.2.1 Material used for fabrication of acoustic device A Mark 4(v)NM shall be as specified on drawings referenced in 2.1.2 with the following exceptions:

3.2.1.1 Components on the following drawings shall be fabricated from corrosion resistant steel AISI type 304:

B-3,103,979	B-3,139,036
B-3,103,980	B-3,139,037
B-3,103,981	B-3,139,042
B-3,103,982	B-3,168,095
B-3,103,983	B-3,168,096
B-3,139,000	B-3,168,097
B-3,139,004	B-3,168,098
B-3,139,006	B-3,168,099
B-3,139,007	B-3,168,100
B-3,139,008	B-3,168,101
B-3,139,010	B-3,168,102
B-3,139,011	B-3,168,103
C-3,139,013	B-3,168,104
B-3,139,014	B-3,168,105
D-3,139,019	B-3,168,106
H-3,139,023	B-3,168,107
H-3,139,024	B-3,168,108
H-3,139,025	B-3,168,110
H-3,139,026	B-3,168,111
B-3,139,027	C-3,168,113
B-3,139,030	D-3,168,114
B-3,139,035	D-3,168,116

3.2.1.2 Components on Drawings B-3,103,997; B-3,103,998; B-3,103,999; B-3,139,002 and B-3,139,018 may be fabricated from any austenitic corrosion resistant steel which maintains a magnetic permeability below 2.0 after fabrication.

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3.2.1.3 Components on Drawing B-3,139,032 shall be fabricated from corrosion resistant steel, AISI type 304 if machined, or class III or MIL-S-17509 if cast.

3.2.2 All fasteners (bolts, nuts and similar items) with the exception of lockwashers shall be fabricated from any austenitic corrosion resistant steel which maintains a magnetic permeability below 2.0 after fabrication. The magnetic permeability of austenitic corrosion resistant steel lockwashers may exceed 2.0.

3.2.3 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used products is allowed under this specification unless otherwise specifically specified.

3.3 Government furnished property. A sample acoustic device, a manual controller, and one copy each of chapter 13, section 13-102 and chapter 23, section 23-102, of NAVSHIPS 250-530, will be supplied the contractor as Government furnished property when the Command or agency concerned is requested by the contractor, via the Government inspector, to furnish such equipment. There will be no other Government furnished property.

3.3.1 The sample device is furnished for general guidance only. It does not necessarily reflect the material, workmanship or dimensional requirements under this specification. The contractor shall be permitted to disassemble the sample device. He shall also be required to reassemble, adjust and test the device for normal operation before returning it to the Government.

3.3.2 The drawings shall not be scaled.

3.3.3 Stock numbers listed on the drawings and in 3.11.1 shall not be used for ordering purposes.

3.4 Construction. Unless otherwise specified herein, the acoustic device shall be constructed in accordance with the drawings.

3.5 Welding. Welding shall be in accordance with MIL-STD-278. The welding electrode used for welding corrosion resistant steel parts shall be type MIL-308 of MIL-E-22200/2. The contractor shall submit the welding procedure, including details of inspection and heat treatment he proposes to follow for approval to the cognizant Government inspector prior to fabrication of the equipment.

3.6 Permeability. Unless otherwise specified herein, all corrosion resistant steel used shall have a magnetic permeability less than 2.0 after fabrication and shall be checked with a permeability indicator conforming to MIL-I-17214.

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3.7 Heat treatment. Assemblies on Drawings H-3,139,023; H-3,139,024; H-3,139,025 and H-3,139,026 for A Mark 4(v)NM acoustic devices shall be heat treated to remove peak stresses for dimensional stability in accordance with MIL-STD-278. Heat treatment for components of the A Mark 4(v) Mod 1 acoustic device shall be in accordance with the requirements of the drawings.

3.8 Identification plates. Identification plates shall be made in accordance with Drawing S8101-B-1,464,365. After painting, the identification plates shall be located as specified on Drawing J-3,136,303.

3.9 "NM" plates. "NM" plates shall be made in accordance with Drawing S8101-A-1,412,161. "NM" plates shall be placed on acoustic device A Mark 4(v)NM only. Before painting, "NM" plates shall be tack welded to acoustic device where shown on Drawing D-3,204,615. Corrosion resistant steel AISI type 304 shall be used for fabricating "NM" plates in lieu of the requirement on the drawing.

3.10 Painting. The diaphragm, electrical contacts, and operating surfaces of the acoustic device shall be unpainted. Exposed internal parts shall be primed in accordance with 3.10.1, 3.10.2, and 3.10.3. Exposed exterior surfaces of streamline housing and sound box shall be painted in accordance with 3.10.1, 3.10.2, 3.10.3 and 3.10.4. The last coat of paint shall be applied after final assembly wherever possible.

3.10.1 The order of operations shall be as follows:

(a) Complete all fabricating operations, such as welding, machining, drilling, and tapping.
(b) Remove all rust and other visible corrosion products.
(c) Remove all grease, oil, and dirt by solvent wiping, vapor degreasing, or caustic washing and rinsing.
(d) Apply pretreatment coating.
(e) Apply primer.
(f) Apply paint.

3.10.2 Pretreatment. One coat of pretreatment primer conforming to DOD-P-15328 shall be applied.

3.10.3 Primer. Two coats of vinyl primer conforming to MIL-P-15930 (except lampblack shall be omitted from the formula) shall be applied.

3.10.4 Paint. Two coats of gray enamel conforming to TT-E-490, color no. 26270 of FED-STD-595, shall be applied as continuous films.

3.11 Repair parts and tools.

3.11.1 On board repair parts and tools. When specified (see 6.2) on board repair parts and tools shall be furnished and shall be in accordance with table I.

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TABLE I. Repair parts and tools.

Description	NAVSEA drawing number	Manufacturer's part number	National stock number	Quantity per vessel	
				On board	Stock
Adjustment bar	B-3,139,000	SPCA B.S. 1025		1	
Bearing, ball, eccentric	H-3,139,028 #13	N.D. 1308	GM3110-155-6705		1
Bearing, ball, pillow	H-3,139,028 #27	N.D. 1306	GM3110-155-6667		2
Belting, "V", adjusting link 36 inches long	H-3,139,028 #8	SPCA B.S. 1062-8	G3030-277-8355	2	1
Bolt, hexagon head, corrosion resistant steel 3/8 inch - 16 NC 2 by 2-3/4 inches long	H-3,139,028 #19		GM5306-227-0889		4
Bolt, hexagon head, CRES (NM) 3/4 inch - 10 NC 3 by 3-1/2 inches long	J-3,136,303 #5		G5306-543-5592	60	10
Bolt, hexagon head, CRES (NM) 5/8 inch - 11 NC 3 by 2-1/2 inches long	J-3,136,303 #10		G5306-543-5589	12	12
Bolt, hexagon head, CRES (NM) 1/2 inch - 13 NC 2 by 1-1/4 inches long	J-3,136,303 #17		GM5306-281-9311	20	20
Bolt, hexagon head, CRES (NM) 1/2 inch - 13 NC 2 by 2-1/2 inches long	J-3,136,303 #15		GM5306-281-9316	4	4
Connector, plug, female	H-3,139,028 #57	SPCA B.S. 1062-57	G5935-197-4967	2	
Connector, plug, male	H-3,139,028 #57	SPCA B.S. 1062-57	G5935-164-0220	2	
Cover plate, eccentric ring R. H.	B-3,103,990	SPCA B.S. 1014	H1075-336-6477		1
Cover plate, eccentric ring L. H.	B-3,103,990	SPCA B.S. 1014	H1075-336-6476		1

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TABLE I. Repair parts and tools. - Continued

Description	NAVSEA drawing number	Manufacturer's part number	National stock number	Quantity per vessel	
				On board	Stock
Cover plate, pillow block	B-3,103,986	SPCA B.S. 1010	H1075-336-6475		2
Diaphragm	D-3,168,115	SPCA B.S. 1086	H1075-332-0038	4	2
Dome assembly	H-3,139,023	SPCA B.S. 1056	H69SPCA50025		1
			1/H1075-394-9733		1
Eccentric ring and ball stud assembly consisting of:			H69SPCA50034		1
1-ball stud	C-3,139,029	SPCA B.S. 1063			
1-ball socket	B-3,103,993	SPCA B.S. 1017			
1-socket ring	B-3,103,992	SPCA B.S. 1016			
1-socket ring nut	B-3,139,003	SPCA B.S. 1028			
1-eccentric ring	B-3,103,987	SPCA B.S. 1011			
Fitting, grease, angle alemite 1/8 inch	H-3,139,028 #69	SPCA B.S. 1062-69	GM4730-172-0031	2	
Fitting, grease, straight alemite 1/8 inch	H-3,139,028 #68	SPCA B.S. 1062-68	GM4730-172-0022	2	
Gasket, diaphragm	D-3,139,017	SPCA B.S. 1050	H5330-291-4069	2	2
Gasket, eccentric ring	B-3,103,985	SPCA B.S. 1008	H5333-297-4992		2
Gasket, pillow block	B-3,103,996	SPCA B.S. 1020	H5330-298-5696		2
Motor repair parts set consisting of:	(Identify motor repair parts set with complete identification plate data for the electric motor used in the parent acoustic device.)			1	1
1-armature					
4-sets brushes					
1-set brush rigging					
1-set bearings					
2-brush holders					
1-set brush rigging with spring					
Nut, self locking, hexagon 3/8 inch - 16NC2	H-3,139,028 #20	ESNA 29U-066	G5310-584-7991	4	4
Nut, hexagon, CRES (NM) 1/2 inch - 13NC3		ESNA	GF5310-584-7992	4	4

See footnote at end of table.

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TABLE I. Repair parts and tools. - Continued

Description	NAVSEA drawing number	Manufacturer's part number	National stock number	Quantity per vessel	
				On board	Stock
Nut, hexagon, CRES (NM) 5/8 inch - 11NC3	J-3,136,303 #20	ESNA	GF5310-584-7990	11	8
Nut, hexagon, CRES (NM) 3/4 inch - 10NC3	J-3,136,303 #4	ESNA	GM5310-584-7993	60	30
Pillow block	D-3,139,012	SPCA B.S. 1043	H3130-140-0730		2
Pin, dowel 5/16 inch diameter by 3/4 inch long	H-3,139,028 #56		G5315-221-5332		4
Pin, dowel 5/16 inch diameter by 1-1/4 inches long	H-3,139,028 #26		G5315-221-5334		4
Plug, pipe, bronze or brass 1/4 inch IPT	J-3,136,303 #29		G4730-187-4238	1	1
Plug, pipe, bronze or brass 3/4 inch IPT	J-3,136,303 #30		G4730-187-4228	1	1
Ring, snap	B-3,103,994	SPCA B.S. 1018	H5340-298-1324		2
Ring, spacing	B-3,103,995	SPCA B.S. 1019	HM69SPCA50000		2
Seal, cover plate	B-3,103,978 #31	SPCA B.S. 1001-31	H5330-290-9985		2
Seal, cover plate eccen-eccentric	B-3,103,978 #17	SPCA B.S. 1001-17	H5330-290-9983		2
Seal, pillow block	B-3,103,978 #30	SPCA B.S. 1001-30	H5330-290-9984		2
Shell assembly	H-3,139,024	SPCA B.S. 1057	H69SPCA-50030		1
			1/ H1075-394-9734		1
Screw, adjusting	B-3,139,004	SPCA B.S. 1030	H5306-275-3808	1	1
Screw, cap, hexagon head 3/8 inch - 16NC2 by 1 inch long	H-3,139,028 #33		GM5305-290-7163		8
Screw, cap SCH 1/4 inch - 20NC2 by 5/8 inch long	H-3,139,028 #44		GM5305-273-5242	2	2

See footnote at end of table.

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TABLE I. Repair parts and tools. - Continued

Description	NAVSEA drawing number	Manufacturer's part number	National stock number	Quantity per vessel	
				On board	Stock
Screw, set, socket head 5/16 inch - 18NC2 by 3/4 inch long	H-3,139,028 #7		GM5305-249-0278	2	2
Shaft, eccentric	D-3,139,016	SPCA B.S. 1048	HM69SPCA50022		2
Spring	D-3,139,022	SPCA B.S. 1055	H1075-292-4092	2	1
Staple	B-3,139,005	SPCA B.S. 1031		2	2
Striker cylinder assembly consisting of:				1	1
1-striker	D-3,139,021	SPCA B.S. 1054	H1075-511-7376		
1-cylinder sleeve	B-3,103,988	SPCA B.S. 1012			
1-SCH cap screw, 3/8 inch - 16NC2 by 1/2 inch long	H-3,139,028 #67		GM5305-273-6507		
1-high collar, lockwasher 3/8 inch	H-3,139,028 #66		G5310-262-3504		
Striker cylinder	D-1,464,495		H69SPCA5006		1
Stud, clamp bar	B-3,103,999	SPCA B.S. 1024	H5307-274-4739	16	16
Stud, motor	B-3,103,998	SPCA B.S. 1022	H5307-274-4740	4	4
Switch, flood assembly consisting of:					
2-shield, switch	B-3,139,009	SPCA B.S. 1040	H1075-346-8956		
2-leader hose	J-3,136,303 #12				
4-regular jam nuts 5/16 - 18NC2	J-3,136,303 #7		GM5310-189-5162		
Tail assembly	H-3,139,025	SPCA B.S. 1058	H69SPCA-50026		1
			1/ H1075-374-9735		1
Washer spring nonlink, high collar 3/8 inch	H-3,139,028 #34				8
Washer, flat, CRES (NM) 1/2 inch	J-3,136,303 #16			24	24
Washer, lock, spring nonlink, 1/2 inch	J-3,136,303 #18		GM5310-637-3794	20	20

See footnote at end of table.

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TABLE I. Repair parts and tools. - Continued

Description	NAVSEA drawing number	Manufacturer's part number	National stock number	Quantity per vessel	
				On board	Stock
Wrench, ring nut	B-3,139,006	SPCA B.S. 1032	H5120-595-9039	1	1
Wrench, strap 18 inches long by 1 inch - 5 inches IPS			G5320-262-8491		1

1/ Stock number for repair part when fabricated from nonmagnetic corrosion resistant steel.

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3.11.2 Per vessel stock repair parts. When specified (see 6.2), per vessel stock repair parts shall be furnished and shall be in accordance with table I.

3.11.3 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.12 Master drawings. When specified (see 6.2), master drawings and certification data sheets for motor shall be furnished in accordance with MIL-M-17413.

3.13 Workmanship. Workmanship shall be of the highest grade throughout; all parts shall be free from rust, fins, burrs, external sharp edges, corners, or surfaces, or any other defects which may impair their serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. Prior to production two first article units consisting of one acoustic device and one set of on board repair parts and tools each (see 3.11) shall be submitted to a Government laboratory for approval. Final acceptance of the first article units shall be made by the Command or agency concerned. First article inspection shall consist of the examination and tests specified in 4.4.1 and 4.4.2.1 through 4.4.2.7. The contractor may acquire all component parts required for use on the contract or order prior to approval of the first article units. Any final assembly work performed by the contractor on the production units prior to approval of the first article units shall be at the contractor's risk.

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4.4 Quality conformance inspection.

4.4.1 Examination. Each device shall be examined to insure correct finish, workmanship, the use of proper material, accurate fit of parts and any other requirements covered by this specification which do not require tests.

4.4.2 Tests. Each device shall be subjected to the tests specified in 4.4.2.1 through 4.4.2.7. The contractor shall furnish the necessary regulated power supply, electric cable, a compressed air supply, a pressure gauge, a fitting to connect the gauge to the drain hole and a 0 to 120 volt direct current (Vdc) voltmeter for the tests.

4.4.2.1 Lubrication. Only grease which is grade 2 of VV-G-632 shall be used. The spring and spring holders shall be covered with above grease before assembling the unit. Over lubrication shall be avoided. The individual bearings are correctly lubricated when the enclosures are about 1/2 full of grease.

4.4.2.2 The devices shall be inspected and adjusted in accordance with the applicable sections of NAVSHIPS 250-530 (see 3.3).

4.4.2.3 The power supply cable shall be connected to the device and the electrical wiring shall be completed in accordance with the wiring diagrams. The direct current (dc) voltage supply shall be checked to make certain the 120 Vdc are available for operation, and that the voltage will remain substantially constant with load fluctuations within plus or minus 5 amperes.

4.4.2.4 The acoustic device shall be started with 120 Vdc nominal starting line voltage and no external field resistance, and the unit shall be operated in air with no internal air pressure for 15 minutes; followed immediately by 15 minutes operation at current of 10 amperes. The line voltage and armature current shall be recorded at 5-minute intervals during the entire test.

WARNING: The full load armature current of the motor is 12-1/2 amperes. Continued operation beyond this rating will cause damage to the motor.

4.4.2.5 The diaphragm shall be removed after the test to determine whether the striker hits the diaphragm squarely. The fly wheel shall be turned several revolutions by hand to determine whether parts of the mechanism assembly move freely.

4.4.2.6 The sound box shall then be filled with air to a pressure of 10 pounds per square inch (lb/in^2) and the entire unit shall be tested for leaks. This shall be accomplished by applying soapy water liberally over the case and then noting the location of any bubbles which may form. Particular attention shall be given to the diaphragm ring, clamping ring, and mounting ring cap.

WARNING: Do not attempt to remove diaphragm or connector from unit while internal pressure exceeds atmospheric. Injury to personnel is apt to occur if this instruction is not adhered to.

4.4.2.7 Each device shall be tested for leaks and electrical continuity.

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4.5 A report of all tests shall be furnished by the contractor to the Government inspector.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Domestic shipment and early equipment installation and for storage of on board repair parts.

5.1.1 Basic equipment or item.

5.1.1.1 Preservation and packaging. Preservation and packaging shall afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the contractor's commercial practice when such meets these requirements.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate and will afford protection against physical or mechanical damage the during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the contractor's commercial practice when such meets these requirements.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, National stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 On board repair parts and tools. Except as specified in 5.1.2.1, on board repair parts and tools shall be preserved and packaged level A (method II), packed level C and marked in accordance with MIL-P-16298 and MIL-E-17555, as applicable.

5.1.2.1 Diaphragm. Each diaphragm shall be coated with preservative type P-4, packaged by method I of MIL-P-116, and packed in style 1 boxes, in the quantities specified (see 6.2), in accordance with PPP-B-621.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation-packaging, packing and marking for shipment shall be as specified (see 6.2).

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5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when acquisition is made.

5.2.1.1 Preservation, packaging and marking.

5.2.1.1.1 Level A.

5.2.1.1.1.1 Acoustic device. The acoustic device shall be preserved and packaged in accordance with method II of MIL-P-116. The sound box shall be utilized as a vapor proof enclosure and the desiccant shall be positioned within the sound box prior to sealing the opening. A warning tag or label indicating that the desiccant shall be removed prior to use of device shall be located in a prominent position on the device.

5.2.1.1.1.2 On board and stock repair parts and tools. Except as specified in 5.2.1.1.1.2.1, repair parts and tools shall be preserved and packaged level A (method II) and marked in accordance with MIL-P-16298 and MIL-E-17555, as applicable.

5.2.1.1.1.2.1 Diaphragms. Each diaphragm shall be coated with preservative type P-4, and packaged by method I of MIL-P-116.

5.2.1.2 Packing.

5.2.1.2.1 Level A and B.

5.2.1.2.1.1 Acoustic device. The acoustic device shall be packed in sheathed crates conforming to MIL-C-104. The selection of the type, class and style of crate shall be at the option of the contractor.

5.2.1.2.2 Repair parts and tools. Except as specified in 5.2.1.2.2.1, repair parts and tools shall be packed level A in accordance with MIL-P-16298 and MIL-E-17555, as applicable.

5.2.1.2.2.1 Diaphragms. Each diaphragm shall be packed in style 2 boxes, in the quantities specified, in accordance with PPP-B-621.

5.2.1.3 Anchoring, blocking, bracing and cushioning. The equipment, repair parts and tools shall be anchored, blocked, braced and cushioned as required to prevent damage from movement during handling, shipment and storage. Anchoring, blocking, bracing and cushioning shall be in accordance with the appendix to MIL-C-104.

5.2.1.4 Marking. In addition to any special marking required by the contract or order, and precautionary marking specified in 5.2.1.1.1.1, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Shipping containers shall have the letters "NM" stencilled on them in 3 inch black letters if equipment contained therein is nonmagnetic.

6. NOTES

6.1 Intended use. The acoustic devices specified herein are intended for use in acoustic minesweeping.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether acoustic device A Mark 4(v) Mod 1 or A Mark 4(v)NM is required.
(c) Whether on board repair parts and tools are required (see 3.11.1).
(d) Whether per-vessel stock repair parts are required (see 3.11.2).
(e) Whether motor master drawings and certification data sheets are required (see 3.12).
(f) Number of diaphragms to be packed in one box (see 5.1.2.1).
(g) Preservation, packaging, packing and marking required if other than those required (see 5.2).

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.3.

6.6 Subject term (key word) listing.

Austenitic
Corrosion resistant steel
Diaphragm
Magnetic permeability

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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